Gold Mining & Exploration

A Complete Guide

Internal Training Material

March 2026

Classification: Internal — All PMH Staff

PARAMOUNT MATTER HOLDINGS

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Reading Guide by Role

PART	TOPIC	ALL	LEGAL	FINANCE	BD	OPS	MGMT
1	**Gold Fundamentals**	●	●	●	●	●	●
2	**Deposit Types**	●	◐	◐	●	●	●
3	**Exploration**	●	◐	●	●	●	●

PART	TOPIC	ALL	LEGAL	FINANCE	BD	OPS	MGMT
4	**Resource & Reporting**	●	●	●	●	◐	●
5	**Mining & Processing**	◐	○	◐	◐	●	◐
6	**Economics & Valuation**	◐	◐	●	●	◐	●
7	**Legal & Regulatory**	◐	●	◐	●	◐	●
8	**Value Chain & Market**	◐	◐	●	●	○	●
9	**Industry Landscape**	○	○	◐	●	○	●
10	**Glossary**	Reference — Use as needed					

● Essential ◐ Recommended ○ Optional

Part 1: Gold Fundamentals ALL

1.1 Why Gold Matters

Gold (chemical symbol: **Au**, from Latin *aurum*) has been valued by humans for over 5,000 years. Understanding its basic physical properties is essential because they directly determine how gold deposits form, how we find gold, and how we extract it.

PROPERTY	VALUE	WHY IT MATTERS
Density	19.3 g/cm³	~7× heavier than rock. This is why gold sinks to the bottom of rivers and why gravity separation works
Chemical Inertness	Does not rust or corrode	Gold survives millions of years of weathering. The rock disappears, but the gold remains
Malleability	Most malleable metal	Gold deforms rather than breaks during river transport — flattened into flakes and nuggets
Melting Point	1,064°C	Achievable with basic equipment, enabling on-site smelting
Crustal Abundance	0.004 ppm	Extremely rare. Even a 1 g/t deposit represents 250× natural concentration

> **What this means for PMH:** Gold's extreme density is the foundation of our Track B alluvial operations — gravity methods work because gold is so much heavier than everything else. Gold's chemical inertness means it survives weathering and accumulates in rivers over millions of years, creating the alluvial deposits we mine in Mali.

1.2 How Gold Is Measured

Grade

Grade is the concentration of gold in rock or sediment — **the single most important number in gold mining**.

UNIT	MEANING	CONTEXT
g/t (grams per tonne)	Grams of gold per tonne of rock	Standard for hard rock mining. 1 g/t = 1 ppm
g/m³ (grams per cubic metre)	Grams of gold per cubic metre of gravel	Standard for alluvial mining
oz/t (ounces per short ton)	Troy ounces per short ton	Older North American reports. 1 oz/t = 34.29 g/t

Grade Benchmarks — Hard Rock

CATEGORY	GRADE RANGE
Earth's average crustal gold	0.004 g/t (background)
Low grade	0.5–1.0 g/t
Medium grade	1.0–5.0 g/t
High grade	5.0–20 g/t
Bonanza grade	>20 g/t

Grade Benchmarks — Alluvial

CATEGORY	GRADE RANGE
Sub-economic	<0.1 g/m³
Low grade	0.1–0.3 g/m³
Medium grade	0.3–0.5 g/m³
Good grade	0.5–1.0 g/m³
High grade	1.0–2.0 g/m³
Very high grade	>2.0 g/m³

Key Unit Conversions

CONVERSION	VALUE
1 troy ounce (oz)	31.1035 grams
1 metric tonne	1,000 kilograms
1 kilogram of gold	32.15 troy ounces
1 short ton	907.185 kilograms

Purity (Fineness)

FINENESS	PURITY	CONTEXT
999.9	99.99% — "four nines"	Investment grade, highest purity
995	99.5%	LBMA Good Delivery minimum
900	90%	Typical doré bar from alluvial operations
750	75% (18 karat)	Common jewellery standard

Part 2: Types of Gold Deposits ALL

2.1 Two Worlds: Hard Rock vs Alluvial

Primary (Hard Rock / Lode) Deposits — Gold locked inside rock, placed there by geological processes millions of years ago. Must be drilled, blasted, crushed, and chemically processed. *This is PMH's Track A business.*

Secondary (Alluvial / Placer) Deposits — Gold freed from rock by natural weathering and concentrated in rivers by gravity. Extracted using simple gravity methods. *This is PMH's Track B business (Mali).*

2.2 Major Hard Rock Deposit Types

TYPE	FORMATION	GRADE	SIZE	KEY EXAMPLES
Orogenic	Hot fluids along deep fault zones during mountain-building	1–20 g/t	<0.5–50+ Moz	Obuasi (Ghana), Loulo (Mali)
Epithermal	Near-surface (<1.5 km), linked to volcanic activity	1–15 g/t	0.5–10 Moz	Hishikari (Japan)
Porphyry	Massive, low-grade, around cooling magma bodies	0.3–1.5 g/t	5–100+ Moz	Grasberg (Indonesia)
Carlin-type	Ultra-fine "invisible" gold in carbonate rocks	1–10 g/t	1–30 Moz	Carlin Trend (Nevada)
VMS	Underwater volcanic centres; gold as by-product	Variable	Variable	—

> **The most important type for PMH is Orogenic** — this is what makes West Africa one of the world's premier gold regions. Nearly all major West African gold mines exploit orogenic deposits hosted in Birimian greenstone belts.

2.3 Alluvial (Placer) Gold Deposits

Alluvial gold forms when primary deposits are weathered and eroded. Gold particles, being 7× heavier than surrounding sand, sink to the bottom of rivers and concentrate at specific points.

Where Gold Concentrates in a River

- **Bedrock surface** — Gold sinks through gravel until it hits solid rock. The bottom 0.5–2.0 m of gravel plus the top 0.3–1.0 m of weathered bedrock is the "pay zone"
- **Bedrock cracks and crevices** — Gold gets trapped in irregularities. Thorough bedrock cleaning is critical
- **Inside of river bends** — Water slows on the inner bank, dropping heavy particles (point bars)
- **Behind obstacles** — Large boulders create low-velocity zones where gold settles
- **Where gradient changes** — Steep stream entering a flat valley; gold drops out as velocity decreases

Types of Secondary Deposits

TYPE	DISTANCE FROM SOURCE	GOLD SHAPE	NOTES
Eluvial (residual)	0–50 m	Angular, rough	Source is directly uphill
Colluvial (slope)	50–500 m	Sub-angular	Transitional
Alluvial (river)	500 m–tens of km	Rounded, flat	Classic placer deposits

2.4 West Africa Gold Geology

West Africa sits on the **West African Craton** — one of Earth's oldest continental cores (>2.0 billion years). The southern **Leo-Man Shield** hosts the **Birimian greenstone belts**, among the most gold-endowed geological formations on Earth.

The key event: the **Eburnean Orogeny** (~2.15–2.07 billion years ago) created deep crustal shear zones that channelled gold-bearing fluids upward, depositing the orogenic gold deposits we mine today.

Major Gold Belts in West Africa

BELT	LOCATION	KEY DEPOSITS	PMH RELEVANCE
Ashanti Belt	Ghana	Obuasi (>30 Moz), Tarkwa	Reference for orogenic gold
Senegalo-Malian Shear Zone	Mali–Senegal	Loulo (~15+ Moz), Fekola (~10+ Moz)	**Direct context for PMH Mali**
Siguiri Basin	Guinea	Siguiri (>10 Moz)	PMH Guinea area
Ity-Bondoukou Belt	Côte d'Ivoire	Ity (>5 Moz), Yaouré (~3 Moz)	PMH CDI licence areas
Burkina Faso Belts	Burkina Faso	Essakane (>10 Moz)	Regional context

Part 3: Exploration — From Idea to Drill Target ALL BD MGMT

3.1 The Exploration Pipeline

Gold exploration follows a systematic funnel: start wide, narrow down progressively. At each stage, you spend more money but reduce uncertainty. The industry average success rate from grassroots to economic discovery is roughly **1 in 500 to 1 in 1,000**.

STAGE	ACTIVITIES	COST	TIMELINE	KEY DELIVERABLE
1. Desktop Study	Review geological maps, satellite imagery, published data	$50K–$500K	1–6 months	Target areas identified
2. Remote Sensing	Satellite analysis, airborne magnetics, gravity	$100K–$2M	3–12 months	Structural maps, anomaly targets
3. Geological Mapping	Field geologists record rocks, structures, alteration	$200K–$1M	2–6 months	Detailed geological maps
4. Geochemistry	Stream sediment, soil sampling (grid), rock chip	$500K–$5M	6–18 months	Gold-in-soil anomalies
5. Ground Geophysics	IP, EM — detect sulfide mineralization through soil	$500K–$3M	3–12 months	Refined drill targets
6. Initial Drilling	Wide-spaced (100–200 m) RC drilling	$2M–$10M	6–18 months	Discovery confirmation
7. Resource Drilling	Infill RC + DD at 25–50 m spacing, 3D modelling	$10M–$50M	1–3 years	JORC/NI 43-101 Resource
8. Feasibility	PEA → PFS → BFS engineering studies	$2M–$30M	1–3 years	Technical/economic viability

Drilling Methods

METHOD	DEPTH	SAMPLE QUALITY	COST	USE
RAB (Rotary Air Blast)	10–50 m	Low	$10–20/m	Reconnaissance, penetrating laterite
RC (Reverse Circulation)	50–300 m	Good	$30–80/m	**Main workhorse for West Africa**
DD (Diamond Drilling)	>2,000 m	Highest (core)	$80–200/m	Detailed geology, metallurgical samples

Fire Assay — The Gold Standard of Analysis

Fire assay has been used for over 500 years to determine gold content:

1. **Sample preparation:** Crush 2–3 kg to <2 mm, split, pulverize to >90% passing 75 µm
2. **Fusion:** 30–50 g sample + flux (PbO, Na_2CO_3, borax) melted at 1,100°C
3. **Cupellation:** Lead button oxidized at 950°C, leaving gold-silver bead
4. **Measurement:** Weighed (gravimetric) or dissolved for AAS/ICP analysis

> **QA/QC is critical:** Every drill program must include blanks (check contamination), certified reference materials (check accuracy), duplicates (check reproducibility), and umpire assays (5–10% sent to a second lab).

The "Nugget Effect"

Gold is distributed extremely unevenly in deposits. A small sample may miss gold entirely, or include one large particle that inflates the result. This is the biggest challenge in estimating alluvial gold grades. **Solution: use large-volume (bulk) samples.**

Part 4: Resource Estimation & Reporting ALL

LEGAL FINANCE

4.1 Resources vs Reserves

This is one of the most important concepts in mining. The distinction has **legal, financial, and technical implications**.

Mineral Resources — Classified by Confidence

CATEGORY	CONFIDENCE	CAN BE USED FOR
Inferred	Low — geologically reasonable but uncertain	PEA only. **Cannot be used in feasibility studies**
Indicated	Moderate — reasonable assumption of continuity	Pre-Feasibility Study (PFS)
Measured	High — known with high confidence	Feasibility Study (FS/BFS)

Mineral Reserves — The Economically Mineable Part

CATEGORY	DERIVED FROM	CONFIDENCE
Probable Reserve	Indicated Resource	Moderate
Proven Reserve	Measured Resource	High

Key principle: Resources are NOT Reserves. Many published Resources will never become Reserves because they fail economic, environmental, or other tests.

4.2 Reporting Standards

STANDARD	JURISDICTION	EXCHANGE	QUALIFIED PROFESSIONAL
JORC	Australia	ASX	Competent Person (CP)
NI 43-101	Canada	TSX, TSX-V	Qualified Person (QP)
S-K 1300	USA	NYSE, NASDAQ	Qualified Person

Legal implications: You **cannot** publicly claim a "resource" or "reserve" without a report signed by a CP/QP under the relevant standard. Using resource/reserve terminology without qualification can constitute securities fraud. A Competent Person must have at least 5 years' relevant experience.

4.3 The Feasibility Ladder

STUDY	ACCURACY	BASED ON	PURPOSE	COST
PEA	±35–50%	Inferred + Indicated	Worth investigating further?	$0.5–2M
PFS	±25–30%	Min. Indicated	Likely economic? Begin permitting	$2–10M
DFS / BFS	±10–15%	Measured + Indicated	Final investment decision; required by banks	$5–30M

Part 5: Mining & Processing Methods ALL OPS

5.1 Open Pit Mining

The most common method in West Africa, especially for near-surface deposits with thick oxidized zones.

PARAMETER	DESCRIPTION	TYPICAL VALUES
Overall Slope Angle	Angle from pit bottom to rim	Laterite: 35–45°, Fresh rock: 45–60°
Bench Height	Height of each step	Small: 5–6 m, Large: 10–15 m
Strip Ratio	Waste : Ore (tonnes)	Economic: 3:1 to 10:1
Min Pit Bottom Width	Space for equipment	30–40 m minimum

Daily Operations Cycle

1. **Drilling** — Blast holes on a grid (2.5–4.0 m spacing)
2. **Blasting** — ANFO or emulsion explosives; strict safety clearance required
3. **Loading** — Excavator loads trucks; grade control directs ore vs waste
4. **Hauling** — Trucks carry ore to crusher, waste to dump
5. **Dewatering** — Continuous pumping, especially critical during rainy season

5.2 Underground Mining

Used when deposits extend too deep for economic open pit mining. Most common method for West African gold:

- **Cut and Fill** — Mine one horizontal slice, backfill, mine next. Best for narrow, high-grade orogenic veins
- **Sub-Level Stoping** — For steep, regular ore bodies; drill and blast from different levels

5.3 Artisanal & Small-Scale Mining (ASM)

Directly relevant to PMH's Track B model. Low mechanization, gravity processing, labour-intensive. Produces ~600–800 tonnes of gold globally per year (15–20% of world mine production).

5.4 Gold Processing — Key Methods

Gravity Separation

EQUIPMENT	HOW IT WORKS	RECOVERY	BEST FOR
Sluice Box	Inclined trough with riffles; gold settles, waste washes away	60–80%	ASM, alluvial
Jig	Pulsating water stratifies by density	70–85%	Medium scale
Spiral	Slurry on spiral; gravity + centrifugal separation	50–70%	No power needed
Shaking Table	Reciprocating deck with riffles and cross-flow water	>95%	Final upgrading
Centrifugal (Knelson)	Enhanced gravity at 60–200G	>95%	Including fine gold

CIL (Carbon-in-Leach) — Industrial Standard

Used for ~80% of global gold production. Ore is crushed, ground to 75 µm, leached with sodium cyanide in agitated tanks while activated carbon adsorbs the dissolved gold. Recovery: **90–96%**. Capital cost: $200–500M+.

Heap Leaching — Low-Cost Alternative

For low-grade (0.3–1.5 g/t), oxidized ores. Cyanide solution percolates through crushed ore stacked on a liner. Recovery: 50–75% at much lower capital cost ($50–150M).

5.5 From Mine to Doré

Borax Smelting

The most important smelting method for small-scale operations:

1. Mix gravity concentrate with borax and soda ash
2. Heat to 1,100–1,200°C
3. Borax forms glassy slag absorbing impurities
4. Gold sinks to bottom
5. Cool, break slag, extract gold button

Output: 85–95% pure doré. Non-toxic, higher recovery than mercury (90%+ vs 30–60%), equipment cost $100–1,000.

Doré Purity by Source

SOURCE TYPE	TYPICAL PURITY
Major hard rock mines	90–95% Au
Mid-tier hard rock mines	75–90% Au
Alluvial mines (PMH)	**80–92% Au**
ASM hand smelting	70–90% Au

Part 6: Mine Economics & Valuation FINANCE BD

MGMT

6.1 Cost Structure

METRIC	DEFINITION	INCLUDES
C1 Cash Cost	Direct cost per ounce	Mining + Processing + Site G&A + Refining + Royalties
AISC	All-In Sustaining Cost	C1 + Sustaining Capex + Corporate G&A + Sustaining Exploration
AIC	All-In Cost	AISC + Growth Capex + Non-sustaining Exploration

> **PMH Advantage:** Our alluvial AISC is estimated at **$800–1,000/oz**, placing us in the **lower quartile** of the global cost curve (median ~$1,400/oz). This means PMH generates strong margins at any gold price.

6.2 Revenue Formula

```
Revenue = Ore processed (t) × Head grade (g/t) × Recovery (%) × Gold price ($/oz) ÷ 31.103
```

6.3 Key Financial Metrics

METRIC	WHAT IT MEANS	MINIMUM THRESHOLD
NPV (Net Present Value)	Present value of all future cash flows minus investment. NPV > 0 = project creates value	Positive, at conservative gold price
IRR (Internal Rate of Return)	Discount rate at which NPV = 0	15–25% (higher for risky jurisdictions)

METRIC	WHAT IT MEANS	MINIMUM THRESHOLD
Payback Period	Time to recover initial investment	Banks want ≤3–5 years

6.4 Valuation Methods

METHOD	BEST FOR	HOW IT WORKS
DCF / NPV	Development to production	Discount all future cash flows to present
EV / Resource oz	Exploration to development	Enterprise Value ÷ Total ounces
Comparable Transactions	Any stage	What did similar projects trade for?
P/NAV	Public companies	Share price ÷ Net Asset Value per share

Typical Valuation Ranges by Stage

STAGE	EV PER RESOURCE OUNCE	P/NAV MULTIPLE
Grassroots exploration	$5–20/oz	—
Advanced exploration	$20–150/oz	0.1–0.3×
Development (feasibility complete)	$100–250/oz	0.3–0.7×
Production	$150–400/oz	0.8–1.2×

> **Key insight for BD:** Value increases dramatically at each de-risking milestone. A JORC resource is worth many times more than drill intercepts. This is the fundamental value creation mechanism in exploration.

6.5 AISC Calculation Example

```
Monthly production:                 2,000 oz Au

Mining cost:                        $320,000
Processing cost:                    $375,000
Site G&A:                           $180,000
Royalty (5% × revenue):             $200,000
Refining fees:                       $10,000
──────────────────────────────────────────────
Total C1 Cash Cost:               $1,085,000  →  $542/oz

Sustaining capex:                   $120,000
Sustaining exploration:              $50,000
──────────────────────────────────────────────
Total AISC:                       $1,255,000  →  $628/oz

At $2,950/oz gold price:
Margin per ounce:                    $2,322
Monthly pre-tax profit:           $4,644,000
```

6.6 Royalties by Country

COUNTRY	ROYALTY	GOV'T FREE CARRY	CORPORATE TAX
Mali	3%	10% + option 10%	25%
Guinea	5%	15%	25%
Cameroon	4%	10%	30%
CDI	3–5% (sliding)	10%	25%
Namibia	3–5%	—	32%
Botswana	3%	15% option	22%

Part 7: Legal & Regulatory Framework LEGAL

MGMT

7.1 Mining Licence Types

In most African jurisdictions, minerals belong to the state. Companies must obtain licences to explore and mine.

LICENCE TYPE	ACTIVITIES	AREA	DURATION
Reconnaissance	Non-invasive surveys	Large (1000s km²)	1–2 years
Exploration Permit	Surveys, sampling, drilling	100–1,000 km²	2–4 years (renewable)
Mining / Exploitation Permit	Full-scale mining	Defined by resource	15–25 years (renewable)

Typical Government Requirements

- **Work commitments:** Minimum exploration expenditure per year
- **Reporting:** Annual technical reports, production data, employment data
- **Environmental:** EIA/ESIA required before mining
- **Local content:** Local hiring and procurement preferences
- **Area relinquishment:** Must give back 25–50% of area at each renewal

7.2 Royalty Structures

TYPE	BASIS	COMMON RATES
Ad Valorem / GRR	% of gross revenue	3–6%
NSR (Net Smelter Return)	% of revenue after smelting/refining costs	1–5%
NPI (Net Profit Interest)	% of net profit after all costs	5–20%

TYPE	BASIS	COMMON RATES
Government Free Carry	Free equity stake in mining company	10–20%

7.3 Joint Venture & Earn-in Agreements

The most common deal structures in exploration-stage mining:

> **Earn-in structure:** Company A (farmer) holds a licence. Company B (farmee) commits to spend $X on exploration over Y years. Upon meeting the expenditure, Company B earns a specified interest (e.g., 51–80%).

Key Contractual Terms

TERM	DEFINITION	WHY IT MATTERS
NSR Royalty	Ongoing royalty on production to original holder	Significantly reduces project economics. Negotiate caps/buybacks
Dilution	What happens if a party doesn't fund its share	Must be precisely defined with formula
Change of Control	What happens if a party is acquired	Can trigger consent or pre-emptive rights
Operatorship	Who manages day-to-day operations	Operator has significant control — ensure approval thresholds
Anti-dilution	Protection below a threshold (usually 10%)	Party typically converts to a royalty
Tail Period	Rights retained after termination	Creates long-term obligations — negotiate carefully

7.4 Environmental & Social Requirements

EIA / ESIA

Required before any mining permit is granted. Includes: baseline environmental studies (12–24 months), impact assessment, mitigation plans, closure plan, public consultation, and independent review.

Social License to Operate

1. **Early, continuous communication** with village chiefs and local government
2. **Prioritize local hiring** — the most effective social licence tool
3. **Fair land compensation** with independent assessors
4. **Community investment:** 1–2% of revenue to community development
5. **Establish a grievance mechanism**

7.5 Country Notes

COUNTRY	KEY POINTS
CDI	Mining Code 2014 (amended 2023). SARL recommended. EP valid 4 yrs, renewable 2×. 10% free carry. Strong French-language legal system
Namibia	Max 4 exploration permits per company. UBO disclosure required. Well-established, investor-friendly framework
Botswana	15% carried interest option. Strong rule of law, one of Africa's most stable
Mali	Code 2019 (amended 2023). Gov't minimum raised to 20%. Royalty 3%. Export permits mandatory. Political risk remains elevated

Part 8: The Gold Value Chain & Market BD

FINANCE

8.1 From Mine to Market

```
Mine Site → On-site Smelting → Doré Bar (80–95% Au)
                                ↓
Refinery (LBMA-certified) → Bullion Bar (≥99.5% Au)
                                ↓
Markets: LBMA London | COMEX | Central Banks | Jewellery | Investment
```

8.2 LBMA Good Delivery Standard

SPECIFICATION	REQUIREMENT
Minimum fineness	995.0 (99.5% pure)
Weight range	350–430 troy ounces (~10.9–13.4 kg)
Marks	Serial number, refiner's stamp, fineness, year

Only ~70 refineries worldwide hold LBMA accreditation. Gold from these facilities is accepted universally. Non-LBMA gold trades at a discount.

8.3 Dubai as a Gold Trading Hub

Dubai (DMCC) is the world's **second-largest physical gold trading hub** after Switzerland.

- 2024 precious metals trade: AED 625 billion (~USD 170 billion)
- 2,500+ precious metals companies in DMCC

- Multiple LBMA-accredited refineries (Kaloti, Emirates Gold, Al Etihad)
- Strategic position between African supply and Asian demand

> **PMH relevance:** Our DMCC Trading Company serves as the entry point for gold into the formal market. Small producers (<1 t/year) typically start with mid-size Dubai traders, then graduate to direct LBMA refinery relationships as compliance documentation matures.

8.4 Gold Pricing

BENCHMARK	DESCRIPTION
LBMA Gold Price	Set twice daily (10:30 AM, 3:00 PM London). The global reference
COMEX Futures	Most actively traded gold futures (CME Group)
Spot Price	Real-time traded price on global markets

8.5 Supply Chain Compliance

LBMA Responsible Gold Guidance (RGG)

All LBMA refineries must conduct due diligence on gold supply chains, based on the **OECD Five-Step Framework**:

1. Establish strong company management systems
2. Identify and assess supply chain risks
3. Design strategy to respond to risks
4. Independent third-party audit
5. Report on due diligence

What PMH Needs

- **Chain of Custody** documentation (mine to buyer)
- **KYC** documentation for all counterparties
- **Certificate of Origin** for all gold exports
- Export permits from host country
- Internal compliance policy aligned with OECD framework

Part 9: Industry Landscape BD MGMT

9.1 Industry Tiers

TIER	MARKET CAP	PRODUCTION	EXAMPLES
Major	>$10B	>1 Moz/year	Newmont, Barrick, Agnico Eagle
Mid-Tier	$2–10B	300K–1M oz	B2Gold, Endeavour, Kinross
Junior Producer	$500M–2B	100K–300K oz	Perseus Mining, Galiano Gold
Junior Explorer	<$500M	Pre-production	—
Private / ASM	—	Variable	**PMH sits here currently**

9.2 Key Players in West Africa

COMPANY	HQ	KEY W. AFRICA ASSETS	PRODUCTION
Barrick Gold	Canada	Loulo-Gounkoto (Mali)	~600K oz/yr from Mali
Newmont	USA	Ahafo, Akyem (Ghana)	~1M oz/yr from Ghana
B2Gold	Canada	Fekola (Mali)	~600K oz/yr
Endeavour Mining	UK	Ity, Lafigué (CDI)	~1.4M oz/yr total
AngloGold Ashanti	South Africa	Siguiri (Guinea)	~3M oz/yr total
Perseus Mining	Australia	Yaouré (CDI)	~500K oz/yr

9.3 Junior Mining Lifecycle

```
Grassroots → Discovery → Resource Definition → Feasibility → Development → Production

Most juniors EXIT via acquisition by a larger company at the development stage,
rather than building and operating the mine themselves.
```

9.4 Funding Sources

SOURCE	STAGE	SIZE	DILUTION
Founder's capital	Grassroots	$0.5–5M	None
Angel / Seed	Early exploration	$1–5M	High
IPO (TSX-V/ASX/AIM)	Resource definition	$5–50M	Moderate-High
Streaming	Development	$50–500M	Non-dilutive (but gives up revenue)
Royalty financing	Development	$10–100M	Non-dilutive (permanent cost)
Project finance	Construction (needs BFS)	$100M–1B+	Non-dilutive (debt)
JV / Earn-in	Exploration	$2–20M expenditure	Gives up project equity

9.5 Market Context (2025–2026)

- **Gold price:** Near all-time highs (~$2,950/oz), driven by central bank buying and geopolitical uncertainty
- **Exploration budgets:** Record levels in 2025, West Africa a primary destination
- **M&A activity:** Majors actively acquiring advanced projects and juniors
- **AI in exploration:** KoBold Metals ($1.2B+ raised) deploying ML for target generation
- **Regulatory tightening:** Mali, Guinea, DRC increasing government stakes and export controls

Part 10: Glossary ALL

TERM	DEFINITION
AISC	All-In Sustaining Cost — total cost per ounce to sustain production
Alluvial	Deposited by water; alluvial gold is found in river sediments and gravel
ASM	Artisanal and Small-Scale Mining
Assay	Laboratory analysis to determine metal content
BFS / DFS	Bankable/Definitive Feasibility Study — final investment decision level
Birimian	Ancient (2.0–2.2 Ga) volcanic-sedimentary rocks in West Africa hosting most gold
Bullion	Refined gold bars meeting LBMA standards (≥99.5%)
C1 Cash Cost	Direct cash cost per ounce (mining + processing + G&A + refining + royalties)
CIL / CIP	Carbon-in-Leach / Carbon-in-Pulp — industrial gold processing methods
CP / QP	Competent Person / Qualified Person — authorized to sign resource reports
Cut-off Grade	Minimum grade for ore to be economic; below this = waste
DCF	Discounted Cash Flow — valuation method
DD	Diamond Drilling — produces rock core, highest quality samples
Dilution	(Mining) Waste mixed with ore. (Corporate) Ownership reduction from new shares
Doré	Semi-pure gold bar (80–95%) produced at mine; needs refining
Earn-in	Earning a project stake by funding exploration expenditure
EIA / ESIA	Environmental (and Social) Impact Assessment
Fire Assay	Standard gold analysis method (fusion at 1,100°C)
Free Carry	Government's free equity in a mining company (10–20%)
Free Gold	Gold recoverable by gravity (not locked in sulfides)
g/t	Grams per tonne — standard hard rock grade unit
g/m³	Grams per cubic metre — standard alluvial grade unit
Greenstone Belt	Belt of metamorphosed volcanic-sedimentary rocks hosting gold
Head Grade	Average gold grade of ore fed into the processing plant
Heap Leach	Low-cost processing: cyanide percolates through crushed ore pile
Inferred Resource	Low-confidence mineral resource; usable only for PEA
Indicated Resource	Moderate-confidence mineral resource; usable for PFS
IP	Induced Polarization — geophysical method detecting sulfides

TERM	DEFINITION
IRR	Internal Rate of Return
JORC	Australian reporting standard for mineral resources/reserves
JV	Joint Venture
Laterite	Deep tropical weathering product covering West African bedrock
LBMA	London Bullion Market Association — sets global gold standard
Measured Resource	High-confidence mineral resource; usable for FS/BFS
Moz	Million ounces
NI 43-101	Canadian reporting standard for mineral resources/reserves
NPV	Net Present Value
NSR	Net Smelter Return royalty
Nugget Effect	Statistical challenge from gold's uneven distribution
Orogenic	Related to mountain-building; main gold deposit type in West Africa
Pay Zone	Layer in alluvial deposit with economic gold grades
PEA	Preliminary Economic Assessment
PFS	Pre-Feasibility Study
Placer	Sand/gravel deposit containing valuable minerals concentrated by gravity
Probable Reserve	Economically mineable part of an Indicated Resource
Proven Reserve	Economically mineable part of a Measured Resource
RC	Reverse Circulation drilling — main workhorse for W. Africa exploration
Recovery Rate	% of gold in ore actually extracted by processing
Refractory Gold	Gold locked inside sulfide minerals; needs pre-treatment
Royalty	Payment to government/licence holder based on production
Shear Zone	Zone of intense crustal deformation; pathway for gold fluids
Strip Ratio	Waste rock : ore ratio
Streaming	Financing: investor buys right to future production at fixed discount
Tailings	Waste material after ore processing
Troy Ounce	31.1035 grams — standard precious metals unit
UBO	Ultimate Beneficial Owner

Appendix: Key Reference Numbers

Current Market Reference (March 2026)

METRIC	VALUE
Gold price (LBMA)	~$2,950/oz
Global AISC median	~$1,400/oz
PMH alluvial AISC (est.)	$800–1,000/oz
PMH daily production (Mali)	~2 kg/day
PMH annualized production	~730 kg ≈ 23,470 oz
PMH annualized revenue potential	~$69M (at $2,950/oz)
Global mine production (2024)	3,661 tonnes
Mali total production (2024)	58.7 tonnes

Unit Conversions

CONVERSION	VALUE
1 troy ounce	31.1035 grams
1 metric tonne	1,000 kilograms
1 short ton	907.185 kilograms
1 oz/t (per short ton)	34.29 g/t
1 kilogram gold	32.15 troy ounces

This guide is prepared by Paramount Matter Holdings for internal training purposes.
It is designed to give every team member a solid foundation in gold mining and exploration.
For questions, contact the PMH management team.